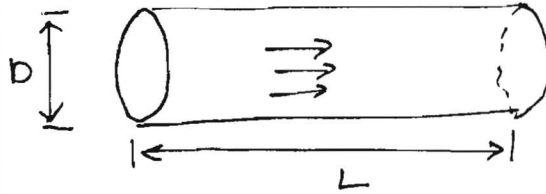




L32 Pressure drop in a pipe flow

Q)

 Δp ? $\Delta p \sim V$? V^2 ? $V^{1.75}$? $\Delta p \sim \frac{1}{D}$? $\frac{1}{D^2}$? $\vdots$

Pumping power = $\Delta p \cdot Q$ ($\because$ 1st law of thermodynamics)
 $\uparrow$
 flow rate [m^3/s]

< Dimensional Analysis >

- laminar flow

$\Delta p = f(u_m, L, D, \mu)$ NOTE: ρ missing why?

$$\frac{D \Delta p}{\mu u_m} = f\left(\frac{L}{D}\right)$$

Since $u_m = \text{const}$, $f\left(\frac{L}{D}\right) = C \cdot \left(\frac{L}{D}\right)$

$$\therefore \Delta p = \frac{C \mu u_m L}{D^2} = \frac{4 C \mu Q L}{\pi D^4}$$

If $C = 32$,

$$\Delta p = \frac{128 \mu L Q}{\pi D^4}$$

Hagen - Poiseuille equation

- turbulent flow

$\frac{\epsilon}{D}$ relative roughness important

ρ density

"



- Darcy friction factor

$$\Delta P = f(u_m, L, D, \mu)$$

↓

$$\frac{\Delta P}{\text{dynamic pressure}} = \frac{\Delta P}{\frac{1}{2} \rho u_m^2} = f(u_m, L, D, \mu, \rho)$$

↓ dimensional analysis

$$\frac{\Delta P}{\frac{1}{2} \rho u_m^2} = f(Re_D, \frac{L}{D})$$

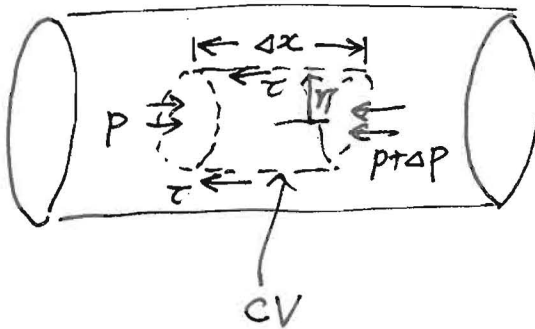
↓

$$\frac{\Delta P / (\frac{\rho}{2} u_m^2)}{\frac{1}{2} \rho u_m^2} = f(Re_D)$$

$$\therefore \boxed{f \equiv \frac{\Delta P}{\frac{1}{2} \rho u_m^2} \cdot \frac{D}{L}}$$

$$\begin{cases} f = f(Re_D) & \text{laminar flow} \\ f = f(Re_D, \frac{\epsilon}{D}) & \text{turb. flow} \end{cases}$$

< Derivation of Hagen - Poiseuille EQ >



Force balance in the fully-developed flow

$$\Delta p \pi r^2 = \tau (2\pi r \Delta x)$$

$$\tau = \frac{1}{2} \Delta p \frac{r}{\Delta x}$$

$$\tau = -\mu \frac{du}{dr} = \frac{1}{2} \left(\frac{dp}{dx} \right) r \quad (\Delta x \rightarrow 0)$$

$$\frac{du}{dr} = -\frac{1}{2} \left(\frac{dp}{dx} \right) \frac{r}{\mu}$$

$$u = -\frac{1}{4} \left(\frac{dp}{dx} \right) \frac{r^2}{\mu} + C_1$$

$$\text{B.C. } u(r = \frac{D}{2}) = 0 \rightarrow C_1 = \frac{1}{16} \left(\frac{dp}{dx} \right) \frac{D^2}{\mu}$$

$$\therefore u(r) = \frac{1}{16} \frac{dp}{dx} \frac{D^2}{\mu} \left\{ 1 - \left(\frac{2r}{D} \right)^2 \right\}$$

$$u(r) = u_0 \left\{ 1 - \left(\frac{2r}{D} \right)^2 \right\}$$

$$u_0 \equiv \frac{1}{16} \left(\frac{dp}{dx} \right) \frac{D^2}{\mu}$$

centerline vel.
max.

$$Q = \int u dA = \int_0^{\frac{D}{2}} u \cdot 2\pi r dr = \frac{1}{128} \frac{\pi D^4}{\mu} \left(\frac{dp}{dx} \right)$$



As $\frac{dP}{dx} = \text{const (why?)}$,

$$\frac{dP}{dx} = \frac{\Delta P}{L} \Rightarrow \Delta P = \frac{128 \mu L Q}{\pi D^4} \quad \text{H.P. EQ}$$

$$f = \frac{64}{Re_D}$$

< Alternative derivation >

Navier-Stokes EQ for a F.D. flow

$$\Rightarrow 0 = -\frac{dP}{dx} + \mu \frac{1}{r} \frac{d}{dr} \left(r \frac{du}{dr} \right)$$

$\Rightarrow$ same results

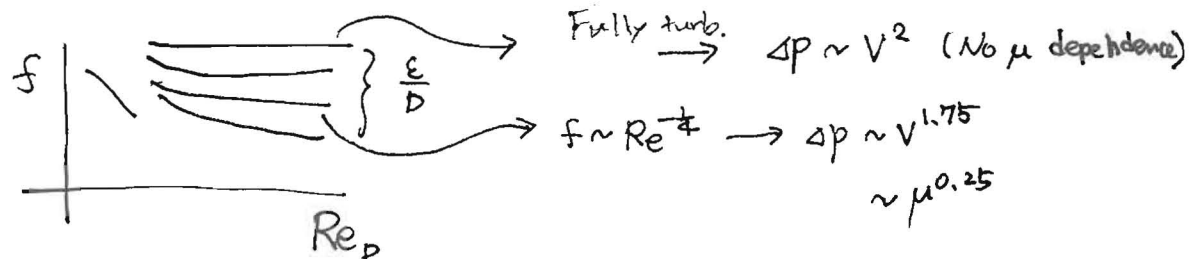
< Pressure drop for general flow >

turbulence $\rightarrow \frac{\epsilon}{D}$, 8 important

general dimensional analysis

$$\Rightarrow f \equiv f(Re_D, \frac{\epsilon}{D})$$

Experimental data $\rightarrow$ Moody chart



$$\frac{1}{\sqrt{f}} = -2.0 \log \left(\frac{\epsilon/b}{3.7} + \frac{2.51}{Re_D \sqrt{f}} \right) \quad \text{turb. flow}$$

$$f = \begin{cases} 0.316 Re_D^{-1/4} & (Re_D \lesssim 2 \times 10^4) \\ 0.184 Re_D^{-1/5} & (Re_D \gtrsim 2 \times 10^4) \end{cases} \quad //$$

$$\frac{1}{\sqrt{f}} = 0.343 \log_{10} Re_D - 1.64 \quad (3000 \lesssim Re_D \lesssim 5 \times 10^6) \quad X$$